Automatic Water-Tank Management System Using FPGA

Verilog HDL | Quartus II | FC-01 Cyclone II FPGA (EP2C8T144C8)

Team Members

Deeraj prasad [Reg. No.513424106705]
Magesh [Reg. No.513424106304]
Madhumitha [Reg. No.513424106033]

Problem Statement

Overflow & water wastage

Manual pump switching often leaves the tank overflowing.

Unnecessary pump running

Pump keeps running even when the tank is already full.

Electricity savings

Pump runs only when water is actually needed.

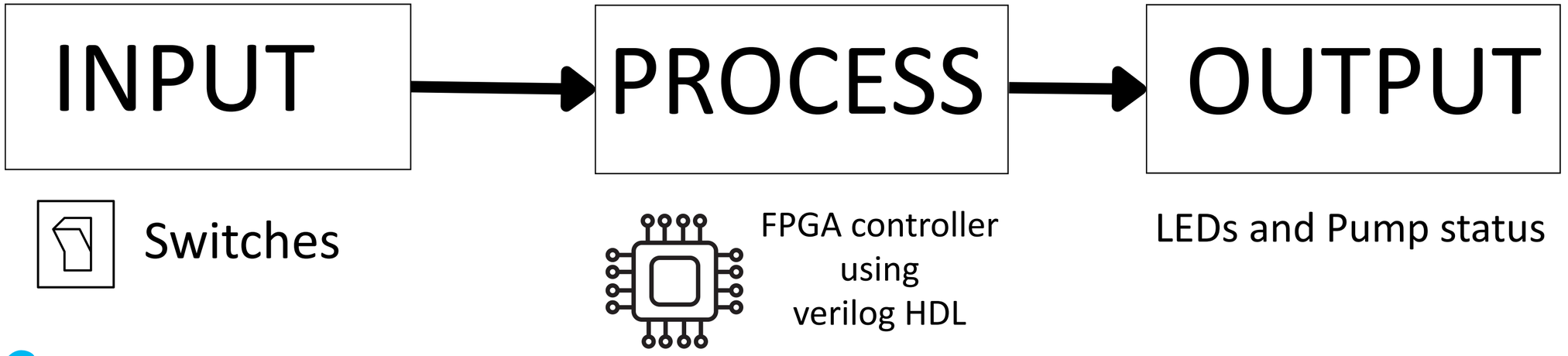
Automatic operation

No one has to keep watching the tank level.

Goal: build an FPGA controller that switches the pump ON at LOW level and OFF at FULL level, automatically.

AIM AND OBJECTIVE

To design and implement a smart control system that efficiently manages resources and integrating user input with logic processing Ultimately enhancing operational awareness and reliability



Working Principle & Block Diagram

Input switches (simulated sensors)

- Switch: LOW sensor
- Switch: MEDIUM sensor
- Switch: HIGH sensor
- Clock + Reset key

FPGA Controller

FC-01 Cyclone II
EP2C8T144C8

FSM: LOW | NORMAL | FULL

Outputs on board LEDs

- LOW LED
- NORMAL LED
- FULL LED
- Pump ON/OFF LED

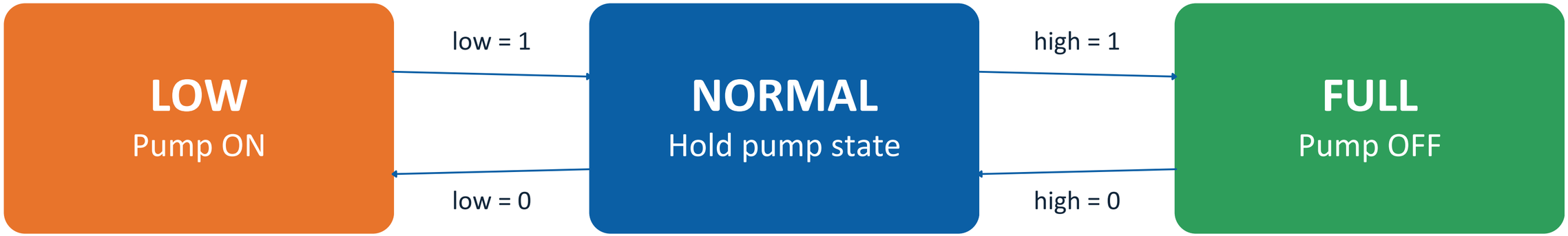
1. Switches simulate the water level (1 = covered, 0 = dry).
2. FSM reads them on each clock edge and picks LOW, NORMAL or FULL.
3. LEDs show the level; the pump LED shows pump ON or OFF.

Truth Table

high med low	Water level	Pump	LOW LED	NORMAL LED	FULL LED
0 0 0	**LOW**	**ON**	ON	OFF	OFF
0 0 1	**NORMAL**	Hold previous	OFF	ON	OFF
0 1 1	**NORMAL**	Hold previous	OFF	ON	OFF
1 1 1	**FULL**	**OFF**	OFF	OFF	ON

1 = sensor covered by water, 0 = dry.
Hold previous: pump stays ON while the level rises from LOW, and stays OFF while it falls from FULL.
*Needs a small validity check in the code (see Future Improvements).

Control Logic (State Machine)



LOW level

Pump ON

NORMAL level

Keep previous pump state

FULL level

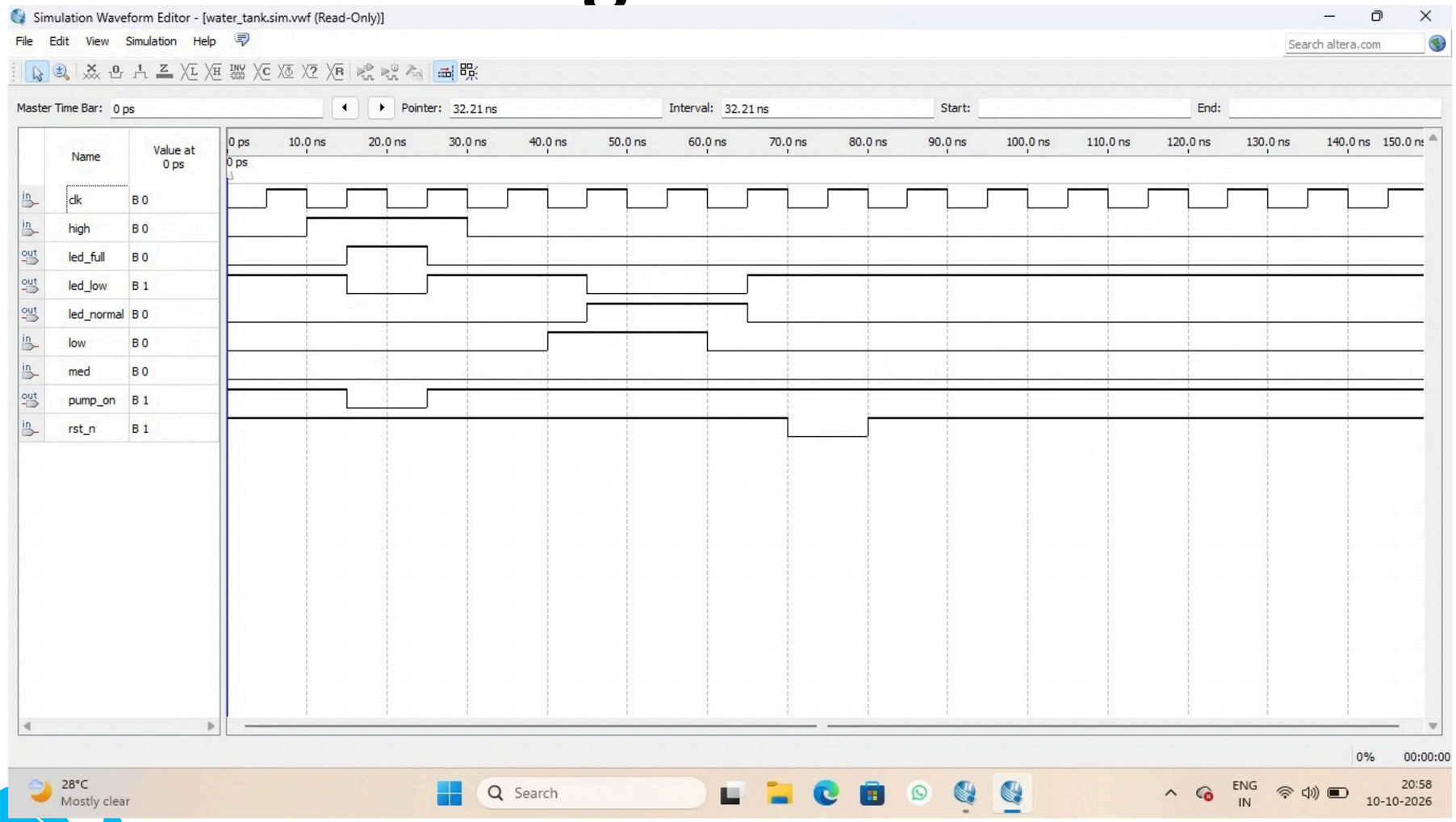
Pump OFF

Invalid input

Safe OFF

On reset the controller starts in the LOW state with the pump ON.

Simulation & Testing



Outputs update on the clock edge after a switch changes. The same steps can be checked in a Quartus waveform simulation.

Advantages, Applications & Future Scope

Advantages

- Low cost, board-only demo
- Fully automatic operation
- Prevents overflow, saves power

Applications

- Homes and apartment buildings
- Laboratories
- Industrial facilities

Limitations

- Switches only simulate sensors
- No real pump driver yet
- Invalid-input check incomplete

Future Improvements

- Real water-level sensors
- Pump driver (relay/contactor)
- Independent overflow protection

THANK YOU